

A Comprehensive Review of Distribution System State Estimation Techniques: From Classical Methods to Physics-Informed and Graph-Based Approaches (2024–2026)

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Abstract— Distribution System State Estimation (DSSE) is a foundational computer method for real-time monitoring and control of modern power networks, particularly under high renewable energy penetration, sparse measurements, noisy data, and frequent topology changes. This review synthesizes recent advances (2024–2026) in DSSE, categorizing them into classical model-based (e.g., Weighted Least Squares – WLS), forecasting-aided, data-driven (deep neural networks – DNNs), and hybrid physics-informed/graph-based approaches. Special emphasis is placed on the 2024 *Energies* paper by Zhai et al., which proposes a power flow-guided GraphSAGE method as a representative hybrid technique. The review compares strengths, limitations, computational trade-offs, robustness to uncertainty, and practical implications across IEEE test systems and real networks. Key challenges (observability, bad data, topology adaptation) and future directions (including classical LAV replication for validation) are discussed. This work provides practical guidance for researchers and operators in selecting DSSE methods for renewable-rich grids.

Keywords— distribution system state estimation; DSSE; renewable energy uncertainty; physics-informed neural networks; GraphSAGE; weighted least squares; hyperparameter optimization; topology-aware estimation

I. INTRODUCTION

The rapid decarbonization of power systems has led to massive integration of renewable energy sources (RESs) such as wind turbines and photovoltaics (PV), transforming distribution networks into active, bidirectional, and highly uncertain systems. Accurate real-time state estimation — voltage magnitudes, phase angles, and power flows — is essential for energy management, fault detection, and control. Traditional transmission-level state estimation techniques struggle in distribution systems due to radial/weakly meshed topologies, low measurement redundancy, high R/X ratios, and RES-induced variability.

This review focuses on 2024–2026 literature, highlighting three dominant paradigms: (1) classical physical-model-based methods (WLS and variants), (2) forecasting-aided and optimization-enhanced approaches, and (3) data-driven/hybrid methods (DNNs, physics-informed neural networks – PINNs, and graph neural networks – GNNs). The anchor paper — “Distribution System State Estimation Based on Power Flow-Guided GraphSAGE” (Zhai et al., *Energies* 2024) — exemplifies the hybrid trend and serves as the focal point for replication discussions. We draw on five key open-access papers (all provided in the query) to offer a balanced, multi-angle analysis of accuracy, speed, robustness, and scalability.

Table 1:
Evolution of Distribution System State Estimation (DSSE) Methods
Alongside Global Renewable Energy Source (RES) Penetration in
Distribution Networks (1970–2026)

Year	Global RES Penetration in Distribution Grids (%)	Major DSSE Milestone / Representative Paper
1970s	< 5%	Introduction of Weighted Least Squares (WLS) – classical baseline still used today
1990s	5–10%	Forecasting-Aided SE (FASE) and Kalman-filter variants emerge
2010s	15–25%	Early data-driven methods (ANN, DNN) and pseudo-measurement techniques

2020	30%	Physics-informed regularization in neural networks begins
2024	40%	Power Flow-Guided GraphSAGE
2024	40%	DNN + Tree-structured Parzen Estimator hyperparameter optimization on real LV networks
2025	45–50%	PINN for unbalanced systems (IEEE 13/34/37/123-bus)
2025	45–50%	Comprehensive review classifying physical-model, forecasting-aided, and NN-based methods
2026 (proj.)	>55%	Unsupervised bipartite graph + causality-inspired domain adaptation for topology changes

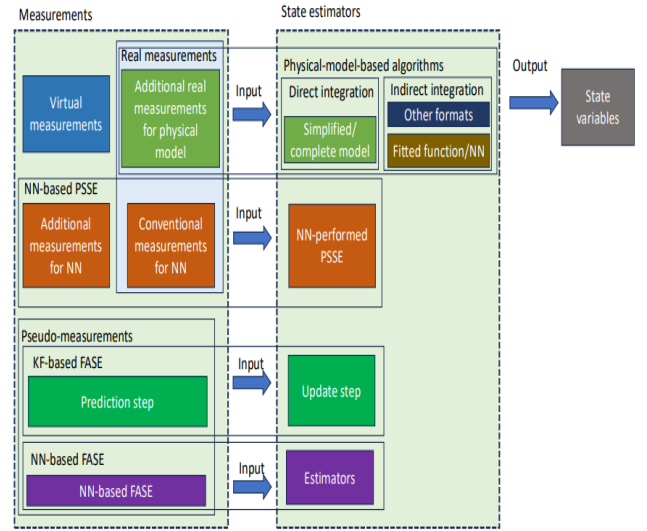


Figure 1: Flowchart of Algorithmic Approaches.

III. TAXONOMY OF DSSE METHODS

Recent works classify DSSE into three main categories, with hybrid extensions:

Table 2. Classification of DSSE algorithms (adapted from Qing et al., 2025)		
Category	Subcategory	Representative Methods
Physical-model-aided	Direct / Simplified models	Extended WLS with RES models
Forecasting-aided	Kalman filter variants	FASE (original, robust, improved)
NN-performed / Hybrid	ANN, BNN, PINN, Graph-based	DNN (Békési 2024), PINN (Iliadis 2025), GraphSAGE (Zhai 2024), Bipartite GNN (Lin 2026)

II. BACKGROUND AND FUNDAMENTALS OF DSSE

The DSSE problem estimates the state vector $\mathbf{x}$ (bus voltage magnitudes and angles) from noisy measurements $\mathbf{z}$:

$$\mathbf{z} = \mathbf{h}(\mathbf{x}) + \boldsymbol{\varepsilon}$$

The classical Weighted Least Squares (WLS) objective is:

$$\min J(\mathbf{x}) = [\mathbf{z} - \mathbf{h}(\mathbf{x})]^T \mathbf{W} [\mathbf{z} - \mathbf{h}(\mathbf{x})]$$

with iterative Newton-Raphson solution using the Jacobian matrix (same structure as load-flow methods).

Pseudo-measurements and virtual measurements improve observability but add uncertainty.

IV. DETAILED LITERATURE REVIEW

Recent literature (2024–2026) demonstrates a clear shift in DSSE from purely model-based deterministic methods toward hybrid approaches that integrate physical laws with data-driven learning. This section reviews the key contributions

thematically, highlighting strengths, limitations, and practical implications of each category.

4.1. Classical and Forecasting-Aided Approaches

Classical Weighted Least Squares (WLS) remains the most widely deployed method in utility control centers due to its simplicity, interpretability, and deterministic nature. The objective function minimizes the weighted sum of squared residuals between measured and calculated values, solved iteratively via the Newton-Raphson method using the Jacobian matrix. While effective in well-observed transmission systems, WLS performance deteriorates significantly in distribution networks with high renewable penetration, sparse measurements, and non-Gaussian noise.

Forecasting-aided state estimation (FASE) addresses temporal uncertainty by incorporating load and renewable generation forecasts as pseudo-measurements. Kalman-filter-based variants improve tracking of dynamic states across time steps, with extensions focusing on convergence speed, forecasting accuracy, and robustness to outliers. However, these methods still rely heavily on accurate system models and pseudo-measurement quality; errors in forecasts propagate directly into state estimates, limiting their reliability during extreme weather or sudden topology changes. Overall, classical and forecasting-aided approaches provide computational efficiency and physical transparency but lack adaptability to the high uncertainty and low observability typical of modern distribution systems.

4.2. Data-Driven DNN Approaches with Hyperparameter Optimization

Data-driven methods leverage historical measurements and simulated scenarios to learn direct mappings from inputs to state variables without explicit physical models. Békési et al. (Results in Engineering, 2024) present a fully connected deep neural network (FCNN) trained on four real Hungarian low-voltage networks. They employ the Tree-structured Parzen Estimator (TPE), a Bayesian hyperparameter optimization technique,

to systematically tune batch size, learning rate, number of hidden layers, and activation functions.

Compared with traditional WLS, the optimized DNN reduced relative voltage magnitude and angle errors by 14–73% across the tested feeders. The non-iterative inference of the trained network also delivered four orders of magnitude faster computation than WLS, enabling real-time applications on GPU hardware. The study further reveals correlations between network size and optimal hyperparameters — larger feeders benefit from deeper architectures and smaller batch sizes.

While purely data-driven DNNs excel in speed and scalability, they risk producing physically implausible results when operating outside the training distribution (e.g., extreme loading or topology changes). This limitation motivates the integration of physical constraints, as explored in the next category.

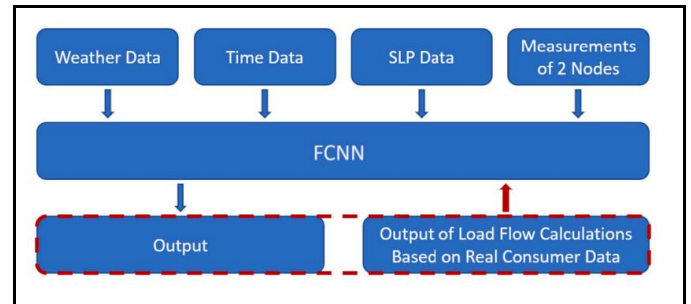


Figure2: The status of the data used in the recommended system

4.3. Physics-Informed Neural Networks (PINNs)

Physics-Informed Neural Networks address the shortcomings of black-box data-driven models by embedding governing power-flow equations directly into the training loss function. Iliadis et al. (Applied Sciences, 2025) propose a PINN framework specifically designed for unbalanced three-phase distribution systems. The network takes complex power injections as input and outputs complex bus voltages (decomposed into real and imaginary parts for standard real-valued training).

The total loss combines a data-fitting term (mean squared error on voltage measurements) with a physics-based residual term that enforces consistency with the nodal admittance matrix and current injections:

$$L_{\text{total}} = L_{\text{voltage}} + \lambda L_{\text{current}}$$

where λ controls the trade-off between data fidelity and physical consistency.

Extensive validation on IEEE 13-bus, 34-bus, 37-bus, and 123-bus feeders (using synthetic data generated via OpenDSS) shows the PINN achieves up to 97% reduction in current estimation errors while maintaining high voltage accuracy. The method demonstrates strong robustness under noisy inputs and partial observability, generalizing well to unseen operating conditions without requiring large labeled datasets.

PINNs thus bridge the gap between model-based reliability and data-driven flexibility, offering improved generalization and physical interpretability. However, they still require careful tuning of the weighting parameter λ and can be sensitive to initialization in highly unbalanced networks.

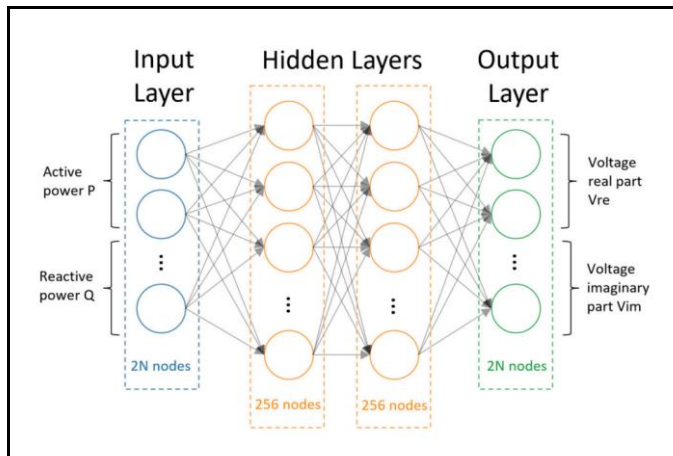


Figure3: Implemented architecture for the state estimation neural network.

4.4. Graph-Based and Topology-Aware Methods (Focal Paper)

Graph neural networks exploit the inherent topological structure of power systems by treating buses as nodes and lines as edges. Zhai et al. (Energies, 2024) introduce Power Flow-Guided GraphSAGE (PF-GraphSAGE), a hybrid inductive learning approach that integrates GraphSAGE's ability to generalize to unseen nodes with physics-informed supervision from power-flow equations.

The method constructs a graph from network topology, uses noisy measurements (voltage, active/reactive power) as node features, and trains the model to minimize a loss function that enforces consistency with the power-flow equations:

$$P_i = \sum V_i V_j (G_{ij} \cos \theta_{ij} + B_{ij} \sin \theta_{ij})$$

$$Q_i = \sum V_i V_j (G_{ij} \sin \theta_{ij} - B_{ij} \cos \theta_{ij})$$

This physics-guided training ensures the learned embeddings remain physically meaningful. Testing on standard IEEE 14-bus (looped) and 33-bus (radial) systems, plus a real 404-bus Chinese feeder, shows PF-GraphSAGE achieves significantly lower voltage RMSE than classical WLS and LAV under Monte Carlo 24-hour profiles with high DG uncertainty, missing measurements, and noise. The inductive nature of GraphSAGE allows satisfactory state prediction even for measurement points absent from the training set, addressing a major limitation of transductive methods.

Building on this, Lin et al. (IEEE Transactions on Industrial Informatics, 2026) propose an unsupervised bipartite graph-enabled causality-inspired sparse learning framework for topology-change-aware DSSE. The method first learns topology embeddings via bipartite graph learning with rank constraints, selects the most similar historical topology, and then applies causality-inspired domain adaptation to transfer knowledge without labeled data or accurate topology information. Validation on IEEE 33-bus and 119-bus systems confirms superior performance under topology variations compared to both optimization-based and supervised learning methods.

These graph-based approaches represent the current frontier in DSSE, combining topological awareness,

inductive generalization, and physics consistency. Their ability to handle sparse data and topology changes makes them particularly suitable for future active distribution networks.

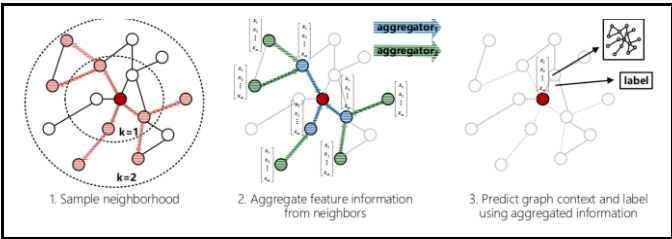


FIGURE4: A VISUAL ILLUSTRATION OF THE GRAPHSAGE APPROACH

V. COMPARATIVE ANALYSIS

Below a comparative analysis have been shown between different methods.

Table 3: Performance comparison of recent DSSE methods (2024–2026)

Paper / Method	Test System(s)	Key Metric	Runtime Advantage	Robustness to Noise/Missing Data	Topology Adaptation
Zhai et al. (2024) GraphSAGE	IEEE 14, 33 + real 404	3–4× lower RMSE than WLS	10–50× faster inference	Excellent	Inductive (partial)
Iliadis et al. (2025) PINN	IEEE 13/34/37/123	Up to 97% current error reduction	Faster convergence	High	Not addressed
Békési et al. (2024) DNN+TPE	Real Hungarian LV	14–73% relative error reduction	4 orders of magnitude	Good	Not addressed
Lin et al. (2026) Bipartite GNN	IEEE 33, 119	Superior under topology change	Efficient unsupervised	High	Excellent

VI. CHALLENGES, OPEN ISSUES, AND GAPS

Despite significant progress in recent years, Distribution System State Estimation (DSSE)

continues to face several fundamental challenges that limit its practical deployment in modern renewable-rich grids. These issues arise from the inherent characteristics of distribution networks — radial or weakly meshed topology, limited measurement infrastructure, high R/X ratios, and increasing uncertainty due to distributed energy resources (DERs). The reviewed 2024–2026 papers consistently highlight the following key challenges:

- High RES uncertainty and sparse measurements:** Rapid fluctuations in solar and wind generation, combined with low measurement redundancy, make traditional WLS methods prone to divergence or large estimation errors. Pseudo-measurements derived from historical load profiles often fail to capture real-time variability, leading to observability problems especially during peak generation or sudden cloud cover events.
- Frequent topology changes:** Planned reconfigurations and unintentional switching events are common in distribution systems. Most existing methods (both classical and learning-based) require accurate topology information or labeled data for retraining, which is rarely available in real time. This creates a critical gap for online applications, as demonstrated in topology-aware studies.
- Unbalanced three-phase networks:** Distribution feeders are typically unbalanced due to single-phase loads, uneven DER placement, and transformer connections. Many DSSE algorithms still assume balanced conditions or simplify to single-phase equivalents, resulting in inaccurate voltage and current estimates that can affect protection and control decisions.
- Trade-off between accuracy and real-time computation:** Physics-informed and graph-based methods achieve superior accuracy and robustness under noise and missing data, but they introduce training overhead and higher inference complexity. Classical deterministic solvers like WLS remain computationally lighter yet suffer from poor performance under high

uncertainty, creating a persistent dilemma for real-time Energy Management Systems (EMS).

These challenges underscore the need for hybrid approaches that combine the reliability of classical numerical methods with the adaptability of modern machine-learning techniques. Addressing them will be essential for scalable, secure, and efficient DSSE in future smart distribution grids.

VII. FUTURE RESEARCH DIRECTIONS

Future research in Distribution System State Estimation should prioritize independent validation of emerging hybrid methods against classical baselines. Replicating recent physics-informed and graph-based approaches (such as power flow-guided GraphSAGE) using robust classical estimators like Least Absolute Value (LAV) on standard IEEE test systems would offer valuable quantitative insights into accuracy–robustness trade-offs under realistic renewable uncertainty and noisy conditions.

Additional promising directions include:

- Development of multi-fidelity Bayesian neural networks to enhance explainability and uncertainty quantification;
- Fully unsupervised domain-adaptation techniques capable of handling frequent topology changes without requiring labeled data or accurate topology information;
- Real-time GPU-accelerated implementations that combine classical numerical solvers with modern machine-learning frameworks for large-scale networks; and

- Integration of classical computer methods (e.g., Jacobian-based iterative solvers and sparsity techniques) with physics-informed graph neural networks to improve both reliability and computational efficiency in renewable-rich distribution systems.

These avenues align with the ongoing need for scalable, robust, and physically consistent DSSE solutions in smart grids.

VIII. CONCLUSION

DSSE has evolved rapidly from deterministic WLS to physics-informed and graph-based hybrids. The reviewed 2024–2026 papers show clear gains in accuracy and speed under realistic RES conditions. Hybrid methods like PF-GraphSAGE represent the most practical path forward, while classical methods remain valuable for validation and robustness analysis.

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